EXAM REVISION 1 — SOLUTIONS
GRADE 10 PAPER 1
MATHEMATICS

QUESTION 1

1.1 $\sqrt{4}\ ;\ 1/3\ ;\ 0\ ;\ \sqrt{\frac{25}{9}}\ ;\ \sqrt[3]{-27}\ ;\ 0,\dot{2}$

1.2.1 INFINITE **1.2.2** $\frac{16}{20}\ ;\ \frac{17}{20} \rightarrow \frac{32}{40}\ ;\ \frac{34}{40} \rightarrow \frac{33}{40}$

1.2.3 $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$ WHICH IS RATIONAL SINCE $ad-bc$ AND bd ARE INTEGERS AND $bd \neq 0$.

1.3.1 FALSE **1.3.2** FALSE **1.3.3** TRUE

1.4.1 $x=1$ **1.4.2** $x=2$ **1.4.3** $x=-1$

1.5.1 LET $x = 0,1232323\ldots \rightarrow 10x = 1,232323\ldots$

AND $1000x = 123,2323\ldots$ $\therefore 990x = 122 \rightarrow x = \frac{122}{990}$

$x = \frac{61}{495}$

1.5.2 LET $x = 0,2333\ldots \rightarrow 10x = 2,333\ldots$ AND $100x = 23,333\ldots$

$\therefore 90x = 21 \rightarrow x = \frac{21}{90} \rightarrow x = \frac{7}{30}$

1.6 $-2\sqrt{7} = -\sqrt{28} \rightarrow \sqrt{25} < \sqrt{28} < \sqrt{36}$

$\therefore 5 < \sqrt{28} < 6 \rightarrow -5 > -\sqrt{28} > -6$

$\therefore -2\sqrt{7}$ LIES BETWEEN -6 AND -5

1.7 $\sqrt[3]{25} = 25^{\frac{1}{3}} = 25^{\frac{2}{6}} = 625^{\frac{1}{6}} = \sqrt[6]{625}$

$2\sqrt{2} = \sqrt{8} = 8^{\frac{1}{2}} = 8^{\frac{3}{6}} = 512^{\frac{1}{6}} = \sqrt[6]{512}$

$\sqrt[6]{625} > \sqrt[6]{512} > \sqrt[6]{500}$ $\therefore \sqrt[3]{25} > 2\sqrt{2} > \sqrt[6]{500}$

1.8.1 $P = 3\left(\frac{6}{2} - 2\right) = 3$ **1.8.2** $P = 3\left(\frac{0}{7} - 2\right) = -6$

1.8.3 $P = 3\left(\frac{x}{x} - 2\right) = -3$

1.9 $(2n+1)(2m+1) = 4mn + 2n + 2m + 1 = 2(mn + n + m) + 1$

WHICH IS ODD (NOTE $m, n \in \mathbb{Z}$, $m \neq n$)

1.10 $\frac{2}{V} = \frac{1}{M} - W \rightarrow \frac{2}{V} = \frac{1-MW}{M} \rightarrow \frac{V}{2} = \frac{M}{1-MV} \rightarrow V = \frac{2M}{1-MV}$

1.11 $\frac{1}{Y} = \frac{1}{x} - \frac{1}{w} \rightarrow \frac{1}{Y} = \frac{w-x}{xw} \rightarrow Y = \frac{xw}{w-x}$, $w \neq x$

1.1.2.1 $P^2 = \frac{1+T}{1-T} \rightarrow P^2 - P^2T = 1+T \rightarrow P^2 - 1 = P^2T + T$

$P^2 - 1 = T(P^2+1) \rightarrow T = \frac{P^2-1}{P^2+1}$

1.1.2.2 LET $T = 1 \rightarrow 1 = \frac{P^2-1}{P^2+1} \rightarrow P^2+1 = P^2-1 \rightarrow 1 = -1$ WHICH IS NOT POSSIBLE

$\therefore T \neq 1$

1.13.1 $\frac{9}{5}C = F - 32 \rightarrow {}^\circ F = \frac{9}{5}{}^\circ C + 32$

1.13.2 $C = \frac{9}{5}C + 32 \rightarrow \frac{4}{5}C = -32 \rightarrow C = -40 \quad \therefore -40^\circ$

1.14.1 $x = -3$ 1.14.2 $x = -3$ 1.14.3 $x = -4$ 1.14.4 $x = 0$

QUESTION 2

2.1.1 $(m+2n)(3m-2n) + 4n^2 = 3m^2 - 2mn + 6mn - 4n^2 + 4n^2$

$= 3m^2 + 4mn$

2.1.2 $\frac{1}{4}(2x-6)(2x-6) = \frac{1}{4}(4x^2 - 24x + 36) = x^2 - 6x + 9$

2.1.3 $\frac{12x^4y}{4xy^3} \times \frac{y^2}{3x^2} - 1 = x - 1$

2.1.4 $\frac{8x+3}{4} - \frac{2x-1}{2} - \frac{5}{4} = \frac{8x+3-2(2x-1)-5}{4}$

$= \frac{8x+3-4x+2-5}{4} = \frac{4x}{4} = x$

2.1.5 $\frac{(x-3)(x+3)}{4(x+3)} \times \frac{2x^2(3x-1)}{2(3x^2-10x+3)} \times 4 = \frac{x^2(3x-1)(x-3)}{(3x-1)(x-3)} = x^2$

2.1.6 $\frac{x}{x-1} - \frac{2}{x+2} - \frac{6}{x^2+x-2} = \frac{x}{x-1} - \frac{2}{x+2} - \frac{6}{(x-1)(x+2)}$

$= \frac{x(x+2)-2(x-1)-6}{(x-1)(x+2)} = \frac{x^2+2x-2x+2-6}{(x-1)(x+2)} = \frac{x^2-4}{(x-1)(x+2)}$

$= \frac{(x-2)(x+2)}{(x-1)(x+2)} = \frac{x-2}{x-1}$

2.1.7 $\left(a^{\frac{1}{2}} \cdot b^{\frac{2}{3}} \cdot c^{\frac{1}{2}}\right)\left(a^{\frac{1}{2}} \cdot b^{-\frac{2}{3}} \cdot c^{\frac{1}{2}}\right) = a \cdot b^0 \cdot c = ac$

2.1.8 $\left(x+\frac{1}{x}\right)\left(\frac{1}{x}+x\right) = 1+x^2+\frac{1}{x^2}+1 = x^2+\frac{1}{x^2}+2$

2.1.9 $\frac{5x^6 \cdot y^{-2}}{15 \cdot x^4 \cdot y^4} \times 3 = \frac{x^2}{y^6}$

2.1.10 $\frac{1}{x}+\frac{1}{y} \times \frac{x^2y^2}{x+y} = \frac{y+x}{xy} \times \frac{x^2y^2}{x+y} = xy$

2.1.11 $\frac{2 \cdot 2(x^2+5x+6)}{x+3} = \frac{4(x+2)(x+3)}{x+3} = 4x+8$

2.1.12 $\frac{(x-2)(x+2)+a(x+2)}{2-x-a} = \frac{(x+2)(x-2+a)}{-(x-2+a)} = -x-2$

2.1.13 $\frac{2(x+5)}{(x+5)(x^2-5x+25)} \times \frac{x^2-5x+25}{2} = 1$

2.1.14 $\frac{5(n-1)}{5n} \times \frac{n^2+n+1}{(n-1)(n^2+n+1)} = \frac{1}{n}$

2.1.15 $\frac{1}{x}+\frac{3}{2}-\frac{1}{3x}=\frac{6+3(3x)-2}{6x}=\frac{9x-4}{6x}$

2.1.16 $\frac{4}{m+2}-4+\frac{10m-2}{2(m+2)}=\frac{8-8(m+2)+10m-2}{2(m+2)}=\frac{6-8m-16+10m}{2(m+2)}$

$=\frac{2m-10}{2(m+2)}=\frac{2(m-5)}{2(m+2)}=\frac{m-5}{m+2}$

2.1.17 $x\neq 1,\ x\neq -2$ 2.1.18 $x\neq -3$

2.2.1 $(x-y)^2=20 \rightarrow x^2-2xy+y^2=20 \rightarrow 10-2xy=20$

$2xy=-10 \rightarrow y=\frac{-5}{x}$

2.2.2 $x\neq 0,\ y\neq 0$

QUESTION 3

3.1.1 $3(a^4-16)=3(a^2-4)(a^2+4)=3(a-2)(a+2)(a^2+4)$

3.1.2 $x^2(1+y)-(y+1)=(y+1)(x^2-1)=(y+1)(x-1)(x+1)$

3.1.3 $2(a+b)^2-(a+b)=(a+b)(2(a+b)-1)=(a+b)(2a+2b-1)$

3.1.4 $3(x^3+8y^3)=3(x+2y)(x^2-2xy+4y^2)$

3.1.5 $x^4(x-1)-(x-1)=(x^4-1)(x-1)=(x^2-1)(x^2+1)(x-1)$

$=(x-1)(x+1)(x^2+1)(x-1)$

3.1.6 $11(a^2-2a-3)=11(a-3)(a+1)$

3.1.7 $x(x^2-4)=x(x-2)(x+2)$

3.1.8 $2(x^2-x-12)=2(x-4)(x+3)$

3.1.9 $9m(m-2) - n(m-2) = (m-2)(9m-n)$

3.1.10 $(x-2-3)(x-2+3) = (x-5)(x+1)$

3.1.11 $\frac{1}{5}(x^3+125) = \frac{1}{5}(x+5)(x^2-5x+25)$

3.1.12 $\frac{1}{2}(x^2-9) = \frac{1}{2}(x-3)(x+3)$

3.2.1 $P = \sqrt{(3n-4)(3n-4)} = \sqrt{(3n-4)^2} = 3n-4$

3.2.2 $3n-4=0 \rightarrow n = 4/3$

3.3.1 $8m^3-27 = (2m-3)(4m^2+6m+9)$ $\therefore AB = 4m^2+6m+9$

3.3.2 $2m-3 \geq 1 \rightarrow 2m \geq 4 \rightarrow m \geq 2$ $\therefore$ MINIMUM: $m=2$

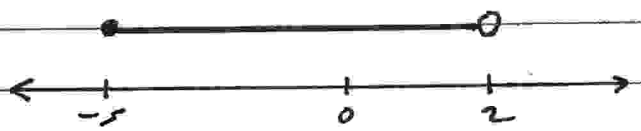
3.3.3 $A(\text{MIN}) = 8.2^3-27 = 37$ UNITS^2

QUESTION 4

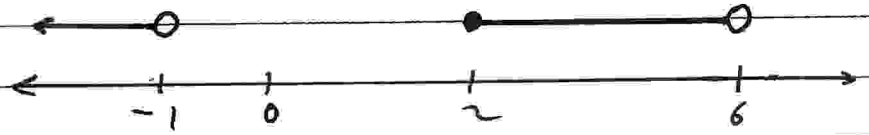
4.1.1 $-4 < -2m \leq 10 \rightarrow -2 < -m \leq 5 \rightarrow 2 > m \geq -5$

$\therefore -5 \leq m < 2$

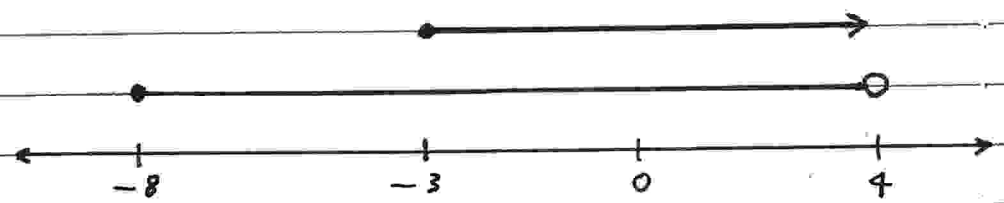
4.1.2



4.2



4.3.1 $-8 \leq x < 4$ AND $x \geq -3$



4.3.2 $x \in [-3; 4)$

4.4.1 $x^2 - x - 12 = 0 \rightarrow (x-4)(x+3) = 0 \rightarrow x = 4$ or $x = -3$

4.4.2 $x^2 - x - 12 \geq x^2 + 2x \rightarrow 3x \leq -12 \rightarrow x \leq -4$

4.5.1 $2x^2 + 5x - 12 = 0 \rightarrow (2x-3)(x+4) = 0 \rightarrow x = \frac{3}{2}$ or $x = -4$

4.5.2 $2x^2 = 15x - 18 \rightarrow 2x^2 - 15x + 18 = 0 \Rightarrow (2x-3)(x-6) = 0$

$\therefore x = \frac{3}{2}$ or $x = 6$

4.5.3 $3x(x-1) = (x+2)(x+2) \rightarrow 3x^2 - 3x = x^2 + 4x + 4$

$\therefore 2x^2 - 7x - 4 = 0 \rightarrow (2x+1)(x-4) = 0 \rightarrow x = -\frac{1}{2}$ or $x = 4$

4.5.4 $\frac{12}{x-3} + \frac{5}{x} = \frac{2}{x^2-3x} \rightarrow \frac{12}{x-3} + \frac{5}{x} = \frac{2}{x(x-3)}$ LCD = $x(x-3)$

$x \neq 0, x \neq 3$

$12x + 5(x-3) = 2 \rightarrow 17x = 17 \rightarrow x = 1$

4.5.5 $3^x + 3^x \cdot 3^2 = 810 \rightarrow 3^x(1+9) = 810 \rightarrow 3^x = 81 \rightarrow 3^x = 3^4 \rightarrow x = 4$

4.5.6 $2^{3x+3} = 64 \rightarrow 2^{3x+3} = 2^6 \rightarrow 3x+3 = 6 \rightarrow 3x = 3 \rightarrow x = 1$

4.5.7 $3^{3x} = 3^2 \cdot 3^{1-x} \rightarrow 3^{3x} = 3^{3-x} \rightarrow 3x = 3 - x \rightarrow 4x = 3 \rightarrow x = 3/4$

4.5.8 $2^{4x-2} = 2^6 \rightarrow 4x - 2 = 6 \rightarrow 4x = 8 \rightarrow x = 2$

4.5.9 $2(2x+1) - 3x = -x + 6$ (LCD = 6)

$4x + 2 - 3x = -x + 6 \rightarrow 2x = 4 \rightarrow x = 2$

4.5.10 $x = \pm\sqrt{m}$ or $x = -n$

4.5.11 $4x^2+4x+1 = x^2-1+1 \rightarrow 3x^2+4x+1=0 \rightarrow (3x+1)(x+1)=0$

$\therefore x = -\frac{1}{3}$ OR $x=-1$

4.5.12 $x^2-20=x \rightarrow x^2-x-20=0 \rightarrow (x-5)(x+4)=0 \rightarrow x=5$ OR $x=-4$

4.5.13 $\frac{5^{3x}+5^{3x}\cdot 5^{-1}}{5^{2x}} = 150 \rightarrow \frac{5^{3x}\left(1+\frac{1}{5}\right)}{5^{2x}} = 150 \rightarrow 5^x = 5^3$

$\therefore x=3$

4.5.14 $\frac{3^{2x}-1}{3^x-1} = 82 \rightarrow \frac{(3^x-1)(3^x+1)}{3^x-1} = 82 \rightarrow 3^x = 3^4 \rightarrow x=4$

4.5.15 $x^2(x+4)-(x+4)=0 \rightarrow (x+4)(x^2-1)=0 \rightarrow (x+4)(x-1)(x+1)=0$

$\therefore x=-4$ OR $x=1$ OR $x=-1$

4.6.1 $2x-y=1 \rightarrow y=2x-1 \rightarrow -2+3(2x-1)=5x$

$-2+6x-3=5x \rightarrow x=5 \rightarrow y=9$

4.6.2 $x+y=15 \rightarrow x=15-y \rightarrow 2^{15-y}=2^{2y} \rightarrow 15-y=2y$

$3y=15 \rightarrow y=5 \rightarrow x=10$

4.6.3 $x+2y=7 \rightarrow 5x+10y=35$ ① AND $5x-3y=-4$ ②

① − ②: $13y=39 \rightarrow y=3 \rightarrow x=1$

4.6.4 $\frac{x}{y}=\frac{2}{3} \rightarrow x=\frac{2y}{3} \rightarrow y+\frac{1}{2}\left(\frac{2y}{3}\right)=4 \rightarrow y+\frac{1}{3}y=4$

$\frac{4}{3}y=4 \rightarrow y=3 \rightarrow x=2$

4.6.5 $4^x\cdot 2^y=1 \rightarrow 2^{2x}\cdot 2^y=1 \rightarrow 2^{2x+y}=2^0 \rightarrow 2x+y=0$ ①

$\frac{7^{y+3}}{49^x}=\frac{1}{7} \rightarrow 7\cdot 7^{y+3}=7^{2x} \rightarrow 7^{y+4}=7^{2x} \rightarrow 2x=y+4$

$\rightarrow 2x-y=4$ ②

① + ② $\rightarrow 4x=4 \rightarrow x=1 \rightarrow y=-2$

4.6.6 $\frac{x-1}{3} = y+2 \rightarrow x-1 = 3y+6 \rightarrow x-3y = 7$ ①

$2x+3 = 8-3y \rightarrow 2x+3y = 5$ ② ①+②: $3x = 12 \rightarrow x = 4$

$\rightarrow y = -1$

4.6.7 $\frac{3}{x} + \frac{6}{y} = 2 \rightarrow \frac{1}{x} + \frac{2}{y} = \frac{2}{3}$ ① $\frac{1}{x} = \frac{1}{y} - \frac{5}{6} \rightarrow \frac{1}{x} - \frac{1}{y} = \frac{5}{6}$ ②

① − ②: $\frac{2}{y} + \frac{1}{y} = \frac{2}{3} + \frac{5}{6} \rightarrow \frac{3}{y} = \frac{9}{6} \rightarrow y = 2 \rightarrow x = -3$

4.7 $3(m-2)+12 > 4(1-m)+24 \rightarrow 3m-6+12 > 4-4m+24$

$\rightarrow m > 22 \rightarrow m > 3\frac{1}{7} \rightarrow m = 4, 5, \ldots, 9$ ∴ 6 SOLUTIONS

4.8.1 $x(m-n) > 5(m-n) \rightarrow x > 5$

4.8.2 $x(m-n) > 5(m-n) \rightarrow x < 5$

(OR) $mx - nx - 5m + 5n > 0 \rightarrow x(m-n) - 5(m-n) > 0 \rightarrow (m-n)(x-5) > 0$

4.8.1 SINCE $(m-n)$ IS +VE, $x-5 > 0 \rightarrow x > 5$

4.8.2 SINCE $(m-n)$ IS −VE, $x-5 < 0 \rightarrow x < 5$

4.9 $(2x-y)(x+3y) = 0 \rightarrow 2x = y$ OR $x = -3y$

$\rightarrow \frac{x}{y} = \frac{1}{2}$ OR $\frac{x}{y} = -3$

4.10 $\frac{10^{2024} - 10^{2022}}{10^{2024} + 10^{2023}} = \frac{10^{2022}(10^2-1)}{10^{2022}(10^2+10)} = \frac{99}{110} = \frac{9}{10}$

4.11 $(x+4)(x+3) = \frac{143}{4} \rightarrow x^2 + 7x + 12 = \frac{143}{4} \rightarrow 4x^2 + 28x - 95 = 0$

$(2x+19)(2x-5) = 0 \rightarrow x = 2, 5$ m

4.12 LET SHAMLA'S AGE $= x$ AND BANJI'S AGE $= y$

THEN $x - 10 = \frac{1}{2}(y-10) \rightarrow x = \frac{1}{2}y + 5$ ①

AND $x + 20 = \frac{6}{7}(y+20) \rightarrow x = \frac{6}{7}y - \frac{20}{7}$ ②

SUBST ① → ②: $\frac{1}{2}Y + 5 = \frac{6}{7}Y - \frac{20}{7} \rightarrow 7Y + 70 = 12Y - 40$

$5Y = 110 \rightarrow Y = 22 \rightarrow X = 16$

∴ SHAMCA IS 16 AND BENTI IS 22

4.13 $\frac{X+1}{Y} = \frac{1}{2} \rightarrow 2X + 2 = Y \rightarrow 2X - Y = -2$ ①

$\frac{X}{Y+3} = \frac{1}{3} \rightarrow 3X = Y + 3 \rightarrow 3X - Y = 3$ ②

② − ①: $X = 5 \rightarrow Y = 12 \rightarrow \frac{X}{Y} = \frac{5}{12}$

4.14 $X + Y = 12$ ① LET Y BE THE LARGER → $Y - 5X = 30$ ②

① − ②: $6X = -18 \rightarrow X = -3 \rightarrow Y = 15$

4.15 $10b + a - (10a + b) = 36 \rightarrow 10b + a - 10a - b = 36$

$9b - 9a = 36 \rightarrow 9(b - a) = 36 \rightarrow b - a = 4$

4.16 $B = 2C$ AND $B = A - 5$ AND $(A + B + C)/3 = 20$

∴ $A + B + C = 60 \rightarrow B + 5 + B + \frac{B}{2} = 60 \rightarrow \frac{5}{2}B = 55 \rightarrow B = 22$

∴ $C = 11$ AND $A = 27$ AND $2029 - 2024 = 5$

∴ CHELSEA WILL BE $11 + 5 = 16$ IN 2029

4.17.1 HOME → DAM: $D = 75t$ AND DAM → HOME: $D = 60\left(t + \frac{3}{4}\right)$

∴ $75t = 60t + 45 \rightarrow 15t = 45 \rightarrow t = 3$ HOURS

∴ $D = 75 \times 3 = 225$ Km (DISTANCE HOME − DAM)

4.17.2 $S = \frac{D}{T} \rightarrow S(\text{AVE}) = \frac{225 \times 2}{3 + 3\frac{3}{4}} \rightarrow$ AVE. SPEED = 66,67 Km/h

4.18 $x - 2 = \frac{15}{2} - \frac{1}{2}Y \rightarrow 2x - 4 = 15 - Y \rightarrow 2x + Y = 19$ ①

$\frac{1}{2}x + 10 = 12 - 3Y \rightarrow x + 20 = 24 - 6Y \rightarrow x = 4 - 6Y$ ②

② → ①: $2(4 - 6Y) + Y = 19 \rightarrow 8 - 11Y = 19 \rightarrow 11Y = -11 \rightarrow Y = -1 \rightarrow X = 10$

4.19 $2x+2y=14 \rightarrow x+y=7$ ① $\quad 3x+6y=30 \rightarrow x+2y=10$ ②

② − ①: $y=3 \rightarrow x=4$

4.20.1 $\frac{R}{x}$ $\quad$ 4.20.2 $\frac{90}{100}x \rightarrow \frac{9}{10}x$

4.20.3 $R\left(\frac{720}{x}+1\right)$

4.20.4 $\frac{9x}{10}\left(\frac{720}{x}+1\right) = 720+144 \rightarrow 648+\frac{9}{10}x = 864$

$\therefore x = \frac{10}{9}(864-648) \rightarrow x = 240$. SUE BOUGHT 240 EGGS

QUESTION 5

5.1.1 $3^4 \cdot 3^{-3} \cdot 3 \rightarrow 3^2 = 9$ $\quad$ 5.1.2 $2^6 \cdot 3^3 \cdot 3^{-2} \cdot 2^{-2} - 1 = 2^4 \cdot 3 - 1$

$= 48 - 1 = 47$

5.1.3 $\frac{(2^2 \cdot 3)^{n+1} \cdot (3^2)^{2n-1}}{(2^2 \cdot 3^2)^n \cdot (2^3)^{1-n}} = \frac{2^{2n+2} \cdot 3^{n+1} \cdot 3^{4n-2}}{2^{2n} \cdot 3^{2n} \cdot 2^{3-3n}}$

$= 2^{2n+2-2n-3+3n} \cdot 3^{n+1+4n-2-2n} = 2^{3n-1} \cdot 3^{3n-1} = 6^{3n-1}$

5.1.4 $\frac{5^n \cdot 5^{-1} + 5^n \cdot 5 + 5^n}{5^n + 5^n \cdot 5^2 + 5^n \cdot 5} = \frac{5^n\left(\frac{1}{5}+5+1\right)}{5^n(1+25+5)} = \frac{\frac{31}{5}}{31} = \frac{1}{5}$

5.1.5 $\frac{2^{2n-2} \cdot 2^{2n+3} \cdot 2^n}{2^n \cdot 2^{3n}} = 2^{4n+3-4n} = 2^3 = 8$

5.1.6 $\frac{5^x \cdot 5 + 7 \cdot 5^x}{5^x \cdot 5 - 5^x} = \frac{5^x(5+7)}{5^x(5-1)} = \frac{12}{4} = 3$

5.1.7(i) $\frac{2^{-n} \cdot 2^{n-2} \cdot 2^{2n+2} \cdot 3^{2n+1}}{3^{2n} \cdot 2^{4n}} = 2^{2n-4n} \cdot 3^{2n+1-2n} = 2^{-2n} \cdot 3$

5.1.7(ii) $2^{-2x} \cdot 3 = 48 \rightarrow 2^{-2x} = 16 \rightarrow 2^{-2x} = 2^4 \rightarrow -2x = 4$

$\therefore x = -2$

5.1.8 $\dfrac{\frac{3}{x}-\frac{2}{y}}{\frac{9y^2-4x^2}{x^2y^2}} = \dfrac{3y-2x}{xy} \times \dfrac{x^2y^2}{(3y-2x)(3y+2x)} = \dfrac{xy}{3y+2x}$

5.1.9.1 $\dfrac{3^{1+x}\cdot 3^{2-x}-3^{3x}}{8\cdot 3^{3x}} = \dfrac{3^{3x}\cdot 3^2-3^{3x}}{8\cdot 3^{3x}} = \dfrac{3^{3x}(9-1)}{3^{3x}\cdot 8} = 1$

5.1.9.2 THE GIVEN EXPRESSION IS INDEPENDENT OF THE VALUE OF x.

5.1.9.3 $\dfrac{81^\circ\cdot 3^2-27^\circ}{8\cdot 3^\circ\cdot 3^\circ\cdot 3^\circ} = \dfrac{1\cdot 9-1}{8\cdot 1} = \dfrac{8}{8} = 1$

5.1.10 $\dfrac{(x+2)(x-1)}{(3x+1)(x-1)} \times \dfrac{(3x+1)(3x+1)}{(2x-3)(x+2)} \times \dfrac{4(2x-3)}{3x+1} = 4$

$\therefore$ INDEPENDENT OF x PROVIDED NO DIVISION BY ZERO.

QUESTION 6

6.1.1 $x+4-(2x+4) = 1-x-(x+4) \rightarrow x+4-2x-4 = 1-x-x-4$

$\therefore x = -3$ $\therefore$ SEQ. IS: $7; -2; 1; 4 \rightarrow d = 3$

$\therefore y = -5$

6.1.2 $T_n = bn+c = 3n+c \rightarrow c = -8 \rightarrow T_n = 3n-8$

(OR) $T_n = a+(n-1)d = -5+3(n-1) \rightarrow T_n = 3n-8$ (CH: $T_4 = 4$ ✓)

6.1.3 $3n-8 > 99 \rightarrow 3n > 107 \rightarrow n > 35\frac{2}{3}$ $\therefore$ THE 36TH TERM ($T_{36} = 100$)

6.1.4 $1025 = 3n-8 \rightarrow 3n = 1033 \rightarrow n = 344\frac{1}{3}$ $\therefore$ 1025 IS NOT A TERM IN THE SEQUENCE

6.1.5 $T_{39} = 3(39)-8 = 109 \rightarrow 109+112+115 = 336$

6.1.6 THE SEQUENCE IS: $-2; 4; 10; 16; \ldots\ldots$ $a = -2$, $d = 6$

$T_n = bn+c \rightarrow T_n = 6n-8 \rightarrow T_{40} = 6(40)-8 = 232$

(OR) $T_n = a+(n-1)d = -2+6(n-1) \rightarrow T_n = 6n-8$ ETC.

6.2.1 $T_{1000} = 2$

6.2.2 000 TERMS : 3 ; 7 ; 11 ; ..... $T_n = b_n + c = 4n - 1$ (CH: $T_4 = 15$ ✓)

$\therefore T_{501}$ IS T_{251} IN 000 TERMS $\rightarrow T_{251} = 4(251) - 1 = 1003$

T_{500} (EVEN) $= 2$ $\therefore T_{501} - T_{500} = 1003 - 2 = 1001$

6.2.3 $4n - 1 < 1000 \rightarrow 4n < 1001 \rightarrow n < 250\frac{1}{4}$ $\therefore n = 250 \rightarrow T_{250} = 999$

6.3.1 67/14 ; 64/17

6.3.2 $T_n = b_n + c$ → NUMERATOR : $T_n = -3n + 82$ $\rightarrow T_n = \dfrac{-3n+82}{3n-1}$

→ DENOMINATOR : $T_n = 3n - 1$

6.3.3 $T_n > 0 \Rightarrow -3n + 82 > 0 \rightarrow 3n < 82 \rightarrow n < 27\frac{1}{3} \rightarrow \therefore$ 27 POSITIVE TERMS

6.3.4 $\dfrac{-3n+82}{3n-1} = \dfrac{-8}{89}$ $\rightarrow -267n + 7298 = -24n + 8$

$243n = 7290 \rightarrow n = 30$

6.3.5.1 $\dfrac{-3n+82}{3n-1} = 1 \rightarrow -3n + 82 = 3n - 1 \rightarrow 6n = 83 \rightarrow n = 13\frac{5}{6}$

$\therefore n = 14$

6.3.5.2 $T_{14} = \dfrac{-3(14)+82}{3(14)-1} = \dfrac{40}{41}$

6.4.1 $x = 1$; $Y = 13$ $(d = 4)$ → (OR) $T_n = a + (n-1)d$

6.4.2 $T_n = b_n + c \rightarrow T_n = 4n - 11$ $= -7 + 4(n-1) = 4n - 11$

6.4.3 $T_{100} = 4(100) - 11 = 389$

6.4.4 $4n - 11 = 869 \rightarrow 4n = 880 \rightarrow n = 220$ (THE 220TH TERM)

6.4.5 $4n - 11 > 999 \rightarrow 4n > 1010 \rightarrow n > 252\frac{1}{2}$ $\therefore n = 253$

6.5.1 $x = -16$ 6.5.2 $T_n = b_n + c \rightarrow T_n = 3n - 28$ (CH: $T_4 = -16$ ✓)

6.5.3 $3n - 28 = 92 \rightarrow 3n = 120 \rightarrow n = 40$ (THE 40TH TERM)

6.5.4 $3n - 28 > 500 \rightarrow 3n > 528 \rightarrow n > 176 \rightarrow T_{177} = 3(177) - 28 = 503$

6.5.5 $\dfrac{3n-28}{3} = n - 9\frac{1}{3}$ WHICH CAN NEVER BE A WHOLE NUMBER.

$\therefore T_n$ IS NOT DIVISIBLE BY 3

6.6.1 3 ; 5 ; 7 ; 9 ∴ 11 DOTS IN PATTERN 5 (↓ CH: $T_5 = 11$ ✓)

6.6.2 $T_n = bn + c \rightarrow T_n = 2n + 1$ OR $T_n = a + (n-1)d = 3 + 2(n-1) = 2n + 1$

6.6.3 $2n + 1 > 550 \rightarrow 2n > 549 \rightarrow n > 274\frac{1}{2} \rightarrow$ ∴ THE 275TH PATTERN

6.6.4 NO. ALL PATTERNS CONTAIN AN ODD NUMBER OF DOTS.

OR $2n + 1 = 6046 \rightarrow 2n = 6045 \rightarrow n = 3022\frac{1}{2}$ ∴ NO

6.7.1 $T_{22} = 3$

6.7.2 ARITHMETIC SEQ: $T_n = bn + c \rightarrow T_n = -4n + 254$. THE 101ST TERM WILL BE THE 51ST TERM $\rightarrow T_{51} = -4(51) + 254 \rightarrow T_{101} = 50$

6.7.3 $-4n + 254 = -146 \rightarrow 4n = 400 \rightarrow n = 100$ (ARITHMETIC TERMS)

∴ THERE IS A TOTAL OF $100 + 99 = 199$ TERMS

6.7.4 $-4n + 254 < 0 \rightarrow 4n > 254 \rightarrow n > 63\frac{1}{2} \rightarrow n = 64$ (ARITHMETIC)

∴ $T_{64} = -4(64) + 254 = -2$ (THE LARGEST NEGATIVE TERM)

QUESTION 7

7.1.1 $d = -1$

7.1.2 $-1 = x - 3 \rightarrow x = 2 \rightarrow Q(2; -1)$

7.1.3 $PQ = 4 \rightarrow P(2; 3) \rightarrow 3 = b^2 - 1 \rightarrow b^2 = 4 \rightarrow b = 2$

∴ $f(x) = 2^x - 1$

7.1.4 $f(0) = 2^0 - 1 = 0 \rightarrow (0; 0) \rightarrow$ f PASSES THROUGH THE ORIGIN

7.1.5 Y-INT: $x = 0 \rightarrow g(0) = -3 \rightarrow$ DISTANCE = 3 UNITS

7.1.6 $x \in (-\infty; 0)$

7.2.1 $n = -2$

7.2.2 x-INT OF g: $y = 0 \rightarrow 0 = -x^2 + 9 \rightarrow x = \pm 3 \rightarrow A(-3; 0)$

∴ $0 = \frac{m}{-3} - 2 \rightarrow m = -6 \rightarrow f(x) = \frac{-6}{x} - 2$

7.2.3 $T'(0; -9)$

7.2.4 $\frac{-6}{x} - 2 = -2x - 6 \rightarrow -6 - 2x = -2x^2 - 6x$

$2x^2 + 4x - 6 = 0 \rightarrow x^2 + 2x - 3 = 0 \rightarrow (x + 3)(x - 1) = 0$

∴ $x = -3$ OR $x = 1$ $f(1) = -6 - 2 = -8$

∴ $(-3; 0)$ AND $(1; -8)$

7.3.1 $f(x) = \frac{q}{x} - 2 \rightarrow -4 = \frac{q}{1} - 2 \rightarrow a = -2 \rightarrow f(x) = \frac{-2}{x} - 2$

7.3.2 x-INT: $y=0 \rightarrow 0 = \frac{-2}{x} - 2 \rightarrow x = -1 \quad \therefore OA = 1$ UNIT

7.3.3 $f(2) = -3 \quad \therefore CD = 1$ UNIT 7.3.4 $y = -x - 2$

7.3.5 $\frac{-2}{x} - 2 = -x - 2 \rightarrow -2 = -x^2 \rightarrow x = \pm\sqrt{2}$

$\therefore x_M = \sqrt{2}$ AND $x_N = -\sqrt{2}$

7.3.6 $y \in \mathbb{R}, y \neq -2$

7.3.7 $f(x) = 3 \rightarrow 3 = \frac{-2}{x} - 2 \rightarrow 5x = -2 \rightarrow x = -\frac{2}{5}$

OR $f(x) = -3 \quad \therefore x = 2$ (FROM 7.3.3)

7.4.1 g: x-INT: $y=0 \rightarrow 0 = x+2 \rightarrow x = -2 \rightarrow A(-2;0)$

$\therefore 0 = 2(-2)^2 + q \rightarrow q = -8$

7.4.2 $2x^2 - 8 = x + 2 \rightarrow 2x^2 - x - 10 = 0 \rightarrow (2x-5)(x+2) = 0$

$\therefore x = 5/2$ OR $x = -2 \rightarrow g(5/2) = \frac{5}{2} + 2 = \frac{9}{2} \rightarrow B\left(\frac{5}{2}; \frac{9}{2}\right)$

7.4.3 $x < -2$ OR $x > 5/2$ 7.4.4 $-2 \leq x \leq 2$

7.4.5 $GH = g(-1) - f(-1) = -1 + 2 - (2 - 8) = 7 \rightarrow GH = 7$ UNITS

7.4.6 $MN = g(x) - f(x) = x + 2 - 2x^2 + 8 \rightarrow MN = -2x^2 + x + 10$

$\therefore 10 = -2x^2 + x + 10 \rightarrow 2x^2 - x = 0 \rightarrow x(2x-1) = 0 \rightarrow x = 0$ OR $x = \frac{1}{2}$

$\therefore AC = 2$ UNITS OR $AC = 2{,}5$ UNITS

7.5.1 $3 = m/2 \rightarrow m = 6$ 7.5.2 $n = 1$

7.5.3 $x + 1 = \frac{6}{x} \rightarrow x^2 + x - 6 = 0 \rightarrow (x+3)(x-2) = 0$

$\therefore x = -3$ OR $x = 2$

$g(-3) = -3 + 1 = -2 \rightarrow A(-3; -2)$

7.5.4.1 $PQ = g(x) - f(x) \rightarrow 2 = x + 1 - 6/x \rightarrow x^2 + x - 6 = 2x$

$x^2 - x - 6 = 0 \rightarrow (x-3)(x+2) = 0 \rightarrow x = 3$ OR $x = -2$

$\therefore OR = 3$ UNITS

7.5.4.2 $f(3) = 2$ AND $g(3) = 4$

$\therefore P(3;4)$ AND $Q(3;2)$

7.5.5 $-4 = \frac{6}{x} \rightarrow x = -\frac{3}{2}$ AND $-4 = x+1 \rightarrow x = -5$

$\therefore CD = 5 - 1,5 = 3,5$ UNITS

7.5.6 $Y = x$ 7.5.7 THEY ARE PARALLEL

7.5.8 $h(x) = \frac{6}{x} + k$ x-INT: $Y = 0 \rightarrow 0 = \frac{6}{x} + k \rightarrow -kx = 6$

$x = -\frac{6}{k} \rightarrow -\frac{6}{k} = \frac{1}{2} \rightarrow k = -12$

(OR) $f(\frac{1}{2}) = 12$ $\therefore$ MUST MOVE DOWN 12 UNITS $\rightarrow k = -12$

7.6.1 $n = 3$

7.6.2 $f(x) = a.m^x + 3 \rightarrow 2 = a.m^0 + 3 \rightarrow a = -1 \rightarrow f(x) = -m^x + 3$

SUB $(1;0) \rightarrow 0 = -m + 3 \rightarrow m = 3 \rightarrow f(x) = -3^x + 3$

7.6.3 $x > 1$ 7.6.4 $Y < 3$

7.6.5 $g(x) = -3^x + \frac{1}{3} \rightarrow 0 = -3^x + \frac{1}{3} \rightarrow 3^x = 3^{-1} \rightarrow x = -1$

7.6.6 $k < -3$

7.7.1 f: x-INT: $Y = 0 \rightarrow 2 = -2/x \rightarrow x = -1$

f: PLOTTING POINT: $x = 2 \rightarrow Y = -3 \rightarrow (2;-3)$

g: Y-INT: $x = 0 \rightarrow Y = -2$; x-INT: $Y = 0 \rightarrow 2 = -x/2 \rightarrow x = -4$

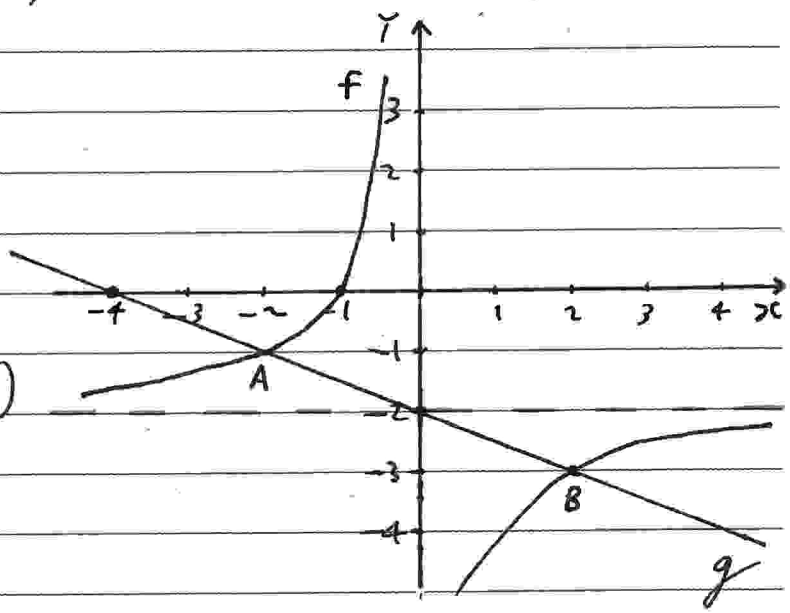
7.7.2 $-\frac{2}{x} - 2 = -\frac{1}{2}x - 2$

$-4 = -x^2 \rightarrow x = \pm 2$

$g(-2) = 1 - 2 = -1$

$g(2) = -1 - 2 = -3$

$\therefore A(-2;-1)$ AND $B(2;-3)$



7.7.3.1 $-1 < x < 0$

7.7.3.2 $-2 \leq x < 0$ OR $x \geq 2$

7.7.3.3 $x = -4$

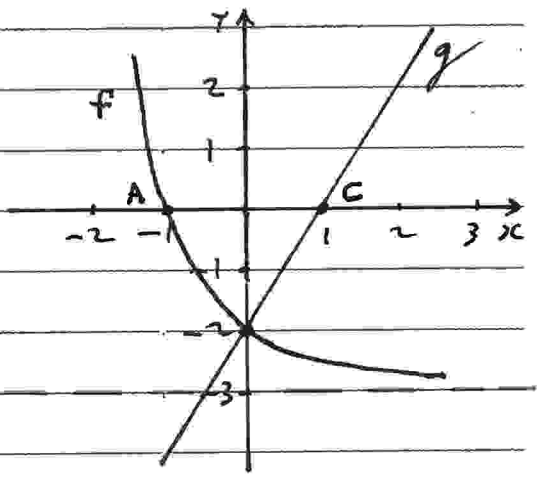
7.8.1 $f(x) = a^x - 3 \rightarrow 6 = a^{-2} - 3 \rightarrow a^{-2} = 9 \rightarrow a^2 = \frac{1}{9} \rightarrow a = \frac{1}{3}$

$\therefore f(x) = \left(\frac{1}{3}\right)^x - 3$

7.8.2 $f(0) = \left(\frac{1}{3}\right)^0 - 3 = 1 - 3 = -2 \quad \therefore K = -2$

7.8.3 f: x-INT: $y = 0 \rightarrow 3 = 3^{-x} \rightarrow 1 = -x \rightarrow x = -1$

g: x-INT: $y = 0 \rightarrow 0 = 2x - 2 \rightarrow x = 1$



7.8.4 AC = 2 UNITS

7.8.5 $-1 < x < 1$

7.9.1 y-INT: $x = 0 \rightarrow f(0) = 4$

x-INT: $y = 0 \rightarrow 0 = 2x + 4 \rightarrow x = -2$

$f(-3) = -6 + 4 = -2$ AND $f(1) = 2 + 4 = 6$

7.9.2 2

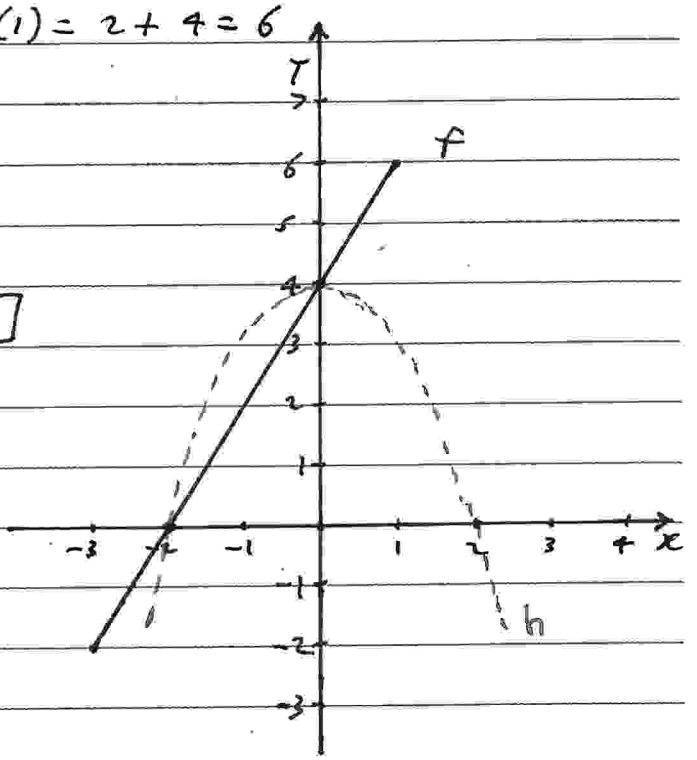
7.9.3 $-\frac{1}{2}$

7.9.4 $-2 \leq y \leq 6$ OR $y \in [-2; 6]$

7.9.5 $h(x) = ax^2 + 4$ SUBST $(-2; 0)$

$0 = a(-2)^2 + 4 \rightarrow 4a = -4$

$\therefore a = -1$ AND $c = 4$



7.9.6 $x \leq 2$

OR $x \in (-\infty; 2]$

$f(x)$	$g(x)$
+	+
−	−
0	ANY
ANY	0

QUESTION 8

8.1.1 10% OF R22000 = R2200 ∴ HP AMOUNT = R19800.00

8.1.2 $A = P(1+n.i) = 19800(1+3,5(0,16)) = R30888.00$

∴ MONTHLY PAYMENTS $= \frac{30888}{42} + 30 = R765.43$

8.1.3 TOTAL PAID = 2200 + 30888 + 42 × 30 = R34348.00

8.2.1 1€ = R18,42638659 → 750€ = R13819.79

1000$ = R17597,35 ∴ TOTAL = R31417.14

8.2.2 1€ = 18,42638659 / 17,59735 $ = 1,047111445 $

∴ 150€ = 157,067 $ → €150 = $157

8.3.1 $A = P(1+i)^n = 25000(1+0,08)^5 = R36733.20$ (AMOUNT IN ACCOUNT)

8.3.2 $A = P(1+ni) \rightarrow 36733.20 = 25000(1+5i) \rightarrow i = 0,0938\ldots$

∴ SIMPLE INTEREST RATE = 9,39% p.a

8.4 $A = P(1+i)^n = 250000(1+0,055)^8 = R383671.63$

∴ COST IS R384000

8.5.1 $A = P(1+ni) \rightarrow 22500 + 14962.50 = 22500(1+x(0,095))$

$\therefore x = 7$

8.5.2 $A = P(1+i)^n \rightarrow 22500 + 14962.50 = 22500(1+i)^7$

$\therefore i = 0,0755\ldots$ ∴ Y = 7,56

8.6.1 $A = P(1+i)^n \rightarrow 7308.46 = x(1+i)^5$ ①

$\rightarrow 11244.98 = x(1+i)^{10}$ ②

① ÷ ②: $1,538625 1 = (1+i)^5 \rightarrow i = 0,0900\ldots$

∴ INTEREST RATE = 9% p.a

8.6.2 $7308.46 = x(1+0,09)^5 \rightarrow x = R4750.00$

8.7.1 $A = P(1+ni)$ $\quad A = 636.32 \times 36 = R\,22907.52$

$\therefore 22907.52 = P(1+3(0,18)) \rightarrow P = R\,14875.01$

$\therefore x = 14875.01 \times \frac{100}{85} = 17500.01 \rightarrow x = 17500$

8.7.2 TOTAL COST $= 22907.52 + 15\%$ OF $17500 = R\,25532.52$

8.7.3 INTEREST $= 22907.52 - 14875.01 = R\,8032.51$

8.8 PETER: $A = P(1+ni) = 2000(1+5(0,09)) = R\,2900.00$

PAULA: $A = P(1+i)^n = 2000(1+0,08)^5 = R\,2938.66$

$\therefore$ PAULA HAD MORE

8.9.1 DEPOSIT $= 7,5\%$ OF $R50000 = R3750.00$

8.9.2 $A = P(1+ni)$ WITH $P = R\,46\,250 \rightarrow A = 46250(1+4(0,17))$

$A = R77700 \rightarrow$ MONTHLY PAYMENTS $= \frac{77700}{48} + 120 = R1738.75$

8.10 $A = P(1+i)^n \rightarrow 24 = 0,13(1+i)^{50} \rightarrow i = 0,1100\ldots$

$\therefore$ ANNUAL INFLATION RATE $= 11\%$ p.a

8.11.1 $1\$ = R\,\frac{1}{0,0512} \rightarrow \$1500 = \frac{1500}{0,0512} = R28846.15 \rightarrow x = 28846.15$

8.11.2 $x = \frac{98,3}{100}\,Y \rightarrow Y = 29345.02$

8.11.3 €75.30 = \$81.324 $= R\,1563.92$

8.12.1 $A = P(1+i)^n = 59,3(1+0,009)^5 = 62,016967\ldots$

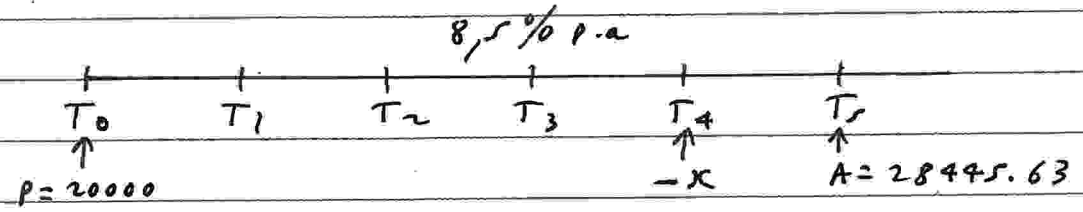
$\therefore$ POPULATION $= 62\,020\,000$

8.12.2 $66,16 = 59,3(1+i)^{10} \rightarrow i = 0,0110\ldots \rightarrow$ RATE $= 1,10\%$ p.a

8.13 $A = P(1+i)^n \rightarrow 3500 = P(1+0,11)^4 \rightarrow P = R2305.56$

$\therefore$ THE ITEM COST R2310 FOUR YEARS AGO.

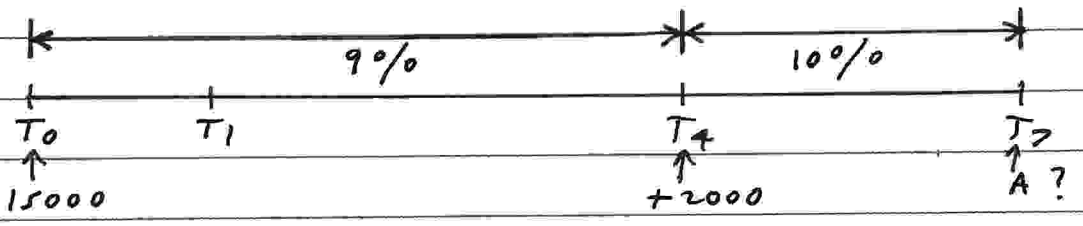
8.14



$A = P(1+i)^n \rightarrow 28445.63 = 20000(1+0,085)^5 - x(1+0,085)$

$\therefore x = 1500$

8.15.1



$A = P(1+i)^n = 15000(1+0,09)^4(1+0,10)^3 + 2000(1+0,10)^3$

$\therefore A = R30844.23$ WHICH IS $< R32000$

8.15.2 $A = 30844.227(1+0,10)^{\frac{1}{2}} = R32349.70$ WHICH IS SUFFICIENT FOR HER HOLIDAY.

QUESTION 9

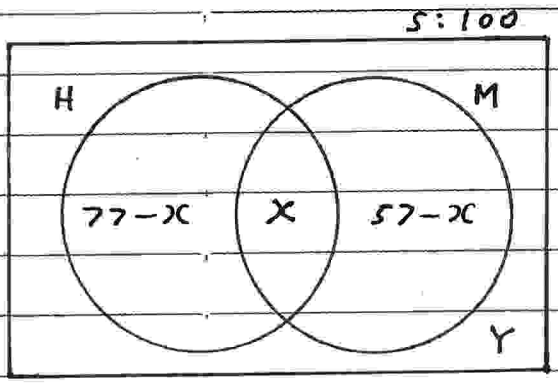
9.1.1 $P(M \text{ OR } N) = P(M) + P(N) - P(M \text{ AND } N) \rightarrow 0,74 = 0,30 + 0,65 - P(M \text{ AND } N)$

$\therefore P(M \text{ AND } N) = 0,21$

9.1.2 NO SINCE $P(M \text{ AND } N) \neq 0$

9.2.1 $\frac{7}{11}$ 9.2.2 $\frac{3}{11}$ 9.2.3 $\frac{2}{11}$

9.3.1



9.3.2 $77 - x + x + 57 - x = 92$

$\therefore x = 42$

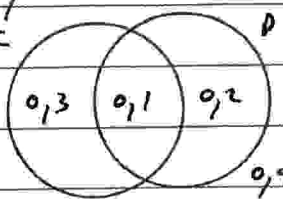
9.3.3 $Y = 8$

9.3.4 $P(H \text{ and } M) = \frac{42}{100} = 0,42$

9.4.1 $P(C) = 0,4$ AND $P(D) = 0,3$

9.4.2 $0,4 - x + x + 0,3 - x = 0,6 \rightarrow x = 0,1$ AND $y = 0,4$

9.4.3 $P(C' \text{ OR } D) = 0,3 + 0,1 + 0,4 = 0,7$



9.5 $A = (x-4)(x+3) = 0 \rightarrow x = 4$ OR $x = -3$

$\therefore P(A=0) = 2/9$

9.6.1.1 NO SINCE $P(A \text{ AND } B) \neq 0$ (OR) NO SINCE A AND B SHARE A COMMON TERM

9.6.1.2 NO SINCE THEY ARE NOT MUTUALLY EXCLUSIVE (OR) NO SINCE $P(A) + P(B) \neq 1$

9.6.1.3 NO SINCE $P(A \text{ OR } B) \neq 1$

9.6.2.1 $n(A) = 6$ 9.6.2.2 $n(A \text{ AND } B) = 1$ 9.6.2.3 $n(\text{NOT } B) = 6$

9.6.3.1 $P(B) = 5/11$ 9.6.3.2 $P(\text{NOT } A) = 5/11$

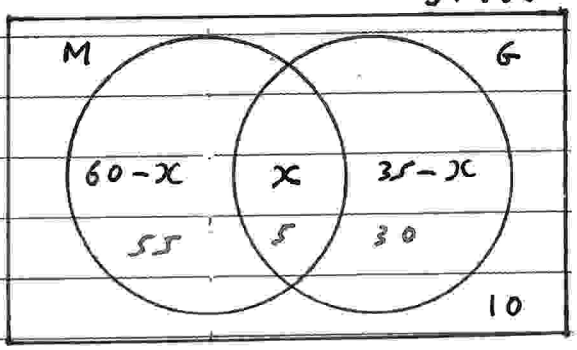
9.6.3.3 $P(A \text{ AND } B) = 1/11$ 9.6.3.4 $P(\text{NOT } A \text{ AND NOT } B) = 1/11$

9.7.1 $4/8 = 1/2$ 9.7.2 $2/8 = 1/4$

9.7.3 $2/8$ 9.7.4 $4/8 = 1/2$

9.8.1

S: 100



9.8.2 $60 - x + x + 35 - x + 10 = 100$

$\therefore x = 5$ (5 PUPILS TAKE BOTH)

9.8.3.1 $P(M \text{ AND } G) = \frac{5}{100} = \frac{1}{20}$

9.8.3.2 $P(M') = \frac{40}{100} = \frac{2}{5}$

9.8.3.3 $P(G \text{ AND } M') = \frac{30}{100} = \frac{3}{10}$

9.9 $-6; -5; -4; -3; -2; -1; 0; 1; 2; 3; 4; 5; 6 \rightarrow 13$

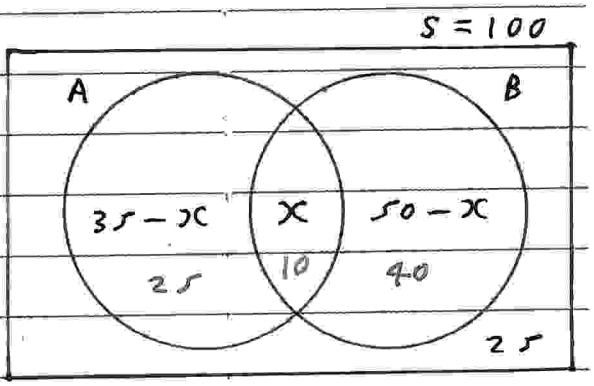
9.9.1 $\frac{2}{13}$ 9.9.2 $\frac{4}{13}$ 9.9.3 $\frac{1}{13}$

9.10.1 $P(E) = \frac{n(E)}{n(S)} \rightarrow \frac{n(A \text{ OR } B)}{n(S)} = \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \text{ AND } B)}{n(S)}$

$\therefore \frac{100-25}{100} = \frac{35}{100} + \frac{50}{100} - \frac{n(A \text{ AND } B)}{100} \rightarrow n(A \text{ AND } B) = 10$

$\therefore$ 10 GIRLS LIKED BOTH SHAMPOOS

9.10.2.1



9.10.2.2 $35 - x + x + 50 - x + 25 = 100$

$\therefore x = 10$ (AS ABOVE)

9.10.3 IN RED $\rightarrow$

9.10.3.1 $P = \frac{10}{100} = \frac{1}{10}$

9.10.3.2 $P = \frac{25}{100} = \frac{1}{4}$ 9.10.3.3 $P = \frac{50}{100} = \frac{1}{2}$

9.10.4 NO, SINCE $P(A \text{ AND } B) \neq 0$

(OR) NO, SINCE EVENTS A AND B SHARE COMMON ELEMENTS